

* Kinetic Energy: "Energy of Motion" $= \frac{1}{2} m v^2 = T$

→ Work: $W = \int_{x_1}^{x_2} \vec{F} \cdot \vec{v} dt = \int_{x_1}^{x_2} \vec{F} \cdot d\vec{x}$

$$\vec{v} = \frac{d\vec{r}}{dt} = x(t)\hat{e}_x + y(t)\hat{e}_y + z(t)\hat{e}_z$$

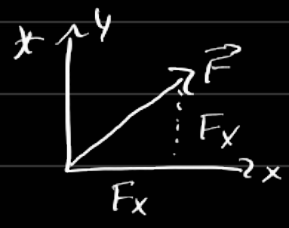
$$d\vec{x} = dx\hat{e}_x + dy\hat{e}_y + dz\hat{e}_z$$

Total Energy:

$$E = T + V$$

* Potential energy: $\vec{F} = -\vec{\nabla} V$

• In 1D: $F = -\frac{dV}{dx} \rightarrow V = -\int F dx$



• Cons. of energy: $T_1 + V_1 = T_2 + V_2$

last time: $F(t) = k \cos(\omega t) \hat{e}_x - mg \hat{e}_y$ *

$$\left\{ \begin{array}{l} \ddot{x} = \frac{k}{m} \cos(\omega t) \quad (F_x) \quad \ddot{y} = -g \quad (F_y) \\ v_x(t) = \frac{k}{m} \sin(\omega t) + \tilde{v}_x \\ x(t) = \frac{k}{m\omega^2} (1 - \cos(\omega t)) + t v_x \end{array} \right. \quad \left\{ \begin{array}{l} v_y(t) = -gt \\ y(t) = -\frac{g}{2} t^2 \end{array} \right.$$

→ Work done by this force

$$\vec{v} = v_x \hat{e}_x + v_y \hat{e}_y \quad | \quad t_1 = 0, \quad t_2 = \frac{2\pi}{\omega}$$

$$W = \int_{t_1}^{t_2} \vec{F} \cdot \vec{v} dt$$

$$\begin{aligned} \vec{F} \cdot \vec{v} &= (k \cos(\omega t) \hat{e}_x - mg \hat{e}_y) \cdot (v_x \hat{e}_x + v_y \hat{e}_y) \\ &= F_x v_x + F_y v_y \\ &= (k \cos(\omega t)) \left(\frac{k}{m} \sin(\omega t) + \tilde{v}_x \right) + (-mg)(-gt) \\ &= \frac{k^2}{m} \cos(\omega t) \sin(\omega t) + \frac{k}{m} \cos(\omega t) \tilde{v}_x + mg^2 t \end{aligned}$$

$$\begin{aligned} W &= \int_{t_1}^{t_2} \left[\left(\frac{k^2}{m} \cos(\omega t) \sin(\omega t) \right) + \left(\frac{k}{m} \cos(\omega t) \tilde{v}_x \right) + mg^2 t \right] dt \\ &= \int_{t_1}^{t_2} \frac{k^2}{m} \cos(\omega t) \sin(\omega t) dt + \int_{t_1}^{t_2} \frac{k}{m} \cos(\omega t) \tilde{v}_x dt + \int_{t_1}^{t_2} mg^2 t dt \end{aligned}$$

① $\frac{k^2}{m} \int_0^{\frac{2\pi}{\omega}} \sin(\omega t) \cos(\omega t) dt = \frac{k^2}{m} \left(-\frac{1}{2\omega} \right) \left[\cos^2(\omega t) \right]_0^{\frac{2\pi}{\omega}}$

another way $u = \sin(\omega t)$

$$du = \omega \cos(\omega t) dt$$

$$\frac{du}{\omega} = \cos(\omega t) dt$$

$$\int \frac{u du}{\omega} = \frac{1}{\omega} \frac{1}{2} u^2 = \frac{1}{2\omega} \sin^2(\omega t) \Big|_0^{\frac{2\pi}{\omega}} = 0$$

$$\cos^2(x) + \sin^2(x) = 1$$

$$\rightarrow \cos(\omega(\frac{2\pi}{\omega})) = 1 \quad 1 - 1 = 0$$

$$\cos(0) = 1$$

$$\sin(\frac{2\pi}{\omega}) = 0$$

$$(1) = 0$$

$$(2) \int_{t_1}^{t_2} \frac{k}{m} \cos(\omega t) \tilde{v}_x dt = \frac{k\tilde{v}_x}{m} \int_{t_1}^{t_2} \cos(\omega t) dt$$

$$= \frac{k\tilde{v}_x}{m\omega} \sin(\omega t) \Big|_0^{2\pi/\omega} = 0$$

$$(2) = 0$$

$$(3) = \int_{t_1}^{t_2} mg^2 t dt = \frac{1}{2} mg^2 t^2 \Big|_0^{2\pi/\omega}$$

$$W = \frac{1}{2} mg^2 \left(\frac{2\pi}{\omega}\right)^2 *$$

* Thm: $\underbrace{\Delta T}_{\text{Kinebe energy}} = W$, $\Delta T = T_2 - T_1$

$\uparrow$ final K.E. $\uparrow$ initial K.E.

$$T_i = \frac{1}{2} m |V_i|^2, \quad \boxed{\vec{V}(0) = \tilde{V}_x \hat{e}_x} \text{ (given as an I.C.)}$$

$$|V_i|^2 = |V_i|^2 = |\vec{V}(0)|^2 = \tilde{V}_x^2$$

$$\vec{V}_f\left(\frac{2\pi}{\omega}\right) = V_x\left(\frac{2\pi}{\omega}\right) \hat{e}_x + V_y\left(\frac{2\pi}{\omega}\right) \hat{e}_y$$

$$= \tilde{V}_x \hat{e}_x + \left(-\frac{2\pi g}{\omega}\right) \hat{e}_y$$

$$|V_f\left(\frac{2\pi}{\omega}\right)|^2 = |V_2|^2 = (\tilde{V}_x)^2 + \left(-\frac{2\pi g}{\omega}\right)^2$$

$$\boxed{|V_2|^2 = \tilde{V}_x^2 + \left(\frac{2\pi g}{\omega}\right)^2}$$

* change in K.E.: $T_2 - T_1$

$$T_2 = \frac{1}{2} m |V_2|^2 = \frac{1}{2} m (\tilde{V}_x^2 + \left(\frac{2\pi g}{\omega}\right)^2) = \frac{1}{2} m \tilde{V}_x^2 + \frac{1}{2} m \left(\frac{2\pi g}{\omega}\right)^2$$

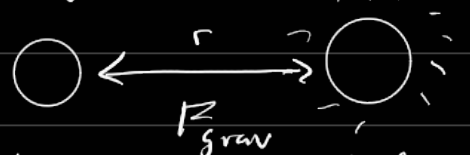
$$T_1 = \frac{1}{2} m \tilde{V}_x^2$$

$$\Delta T = \frac{1}{2} m \tilde{V}_x^2 + \frac{1}{2} m \left(\frac{2\pi g}{\omega}\right)^2 - \frac{1}{2} m \tilde{V}_x^2$$

$$= \frac{1}{2} m \left(\frac{2\pi g}{\omega}\right)^2$$

$$\boxed{\Delta T = \frac{2\pi^2 g^2 m}{\omega^2}} *$$

$$F_{\text{grav}} \propto \frac{1}{r^2}$$



ex: Gravity is a central force and these types of forces

are also called conservative forces: $F = -\frac{dV}{dx} *$

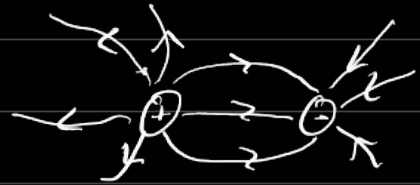
* Not all forces can be written this way. only conservative forces can.

• ex: Electric field: F_{elec} is also

central $\Rightarrow$ conservative force

$$F_{elec} = \frac{q_1 q_2}{4\pi\epsilon_0 r^2} \quad q_i: \text{charges}$$

$$\propto \frac{1}{r^2}$$



* Look at $F = \frac{\alpha}{x^2}$, $\alpha: \text{const.}$

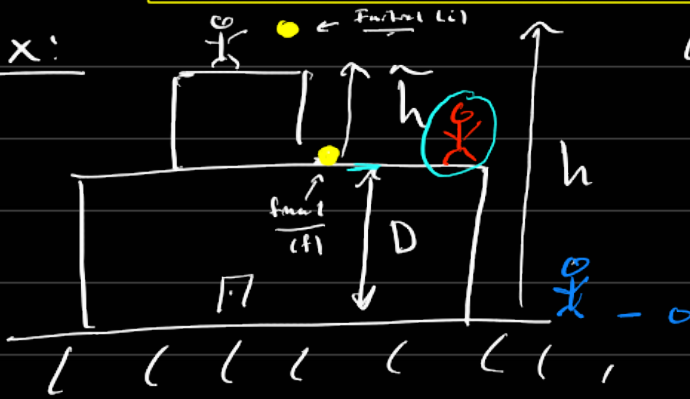
Potential?:

$$V(x) = \frac{\alpha}{x} + C$$

generally that means $C=0$

C : we can set this to any value based on the problem.

ex:



Case (I): measure from ground
 h

Case (II): measure from "root"
 $\tilde{h}$

$$D = h - \tilde{h}$$

Case (1): Ground is zero

Case (2): the "root" is zero

Conservation of energy $T_1 + V_1 = T_2 + V_2$

$$T_1 = \frac{1}{2}mv_i^2 = 0, \quad T_2 = \frac{1}{2}mv_f^2$$

$\rightarrow$ Differs in both cases for the potential

Case (1): $V_1 = mgh$, $V_2 = mgD$

Case (2): $V_1 = mg\tilde{h}$, $V_2 = mg(0) = 0$

Cons. of energy

Case (1): $T_1 + V_1 = T_2 + V_2 \Rightarrow mgh = \frac{1}{2}mv_f^2 + mg(D)$

$$mgh = \frac{1}{2}mv_f^2 + mg(h - \tilde{h})$$

$$= \frac{1}{2}mv_f^2 + mgh - mg\tilde{h}$$

$$0 = \frac{1}{2}mv_f^2 - mg\tilde{h}$$

$$D = h - \tilde{h}$$

$$mgh = \frac{1}{2}mv_f^2$$

Case (2) : $T_1 + V_1 = T_2 + V_2$

$$0 + mgh = \frac{1}{2}mv_f^2 + 0$$

$$mgh = \frac{1}{2}mv_f^2$$

